

PHOENIX_ENGINE_STATUS

Visão Geral

Phoenix Engine é uma game engine 3D educacional construída em C++ e OpenGL.

Objetivos atuais:

Aprender arquitetura de engines.

Construir sistemas do zero.

Priorizar entendimento antes de abstrações.

Inicialmente atingir qualidade visual próxima à era PS2.

Evoluir gradualmente para recursos gráficos mais avançados.

Sistemas Implementados

Core

- ☐ Application
- ☐ Renderer
- ☐ Scene
- ☐ GameObject
- ☐ Component System

Transform

- ☐ Position
- ☐ Rotation
- ☐ Scale
- ☐ Transform Matrix

Camera

- ☐ View Matrix
- ☐ Projection Matrix
- ☐ ViewProjection Matrix

Rendering

- ☐ OpenGL Initialization
- ☐ Shader Compilation
- ☐ Vertex Array Object (VAO)
- ☐ Vertex Buffer Object (VBO)
- ☐ Element Buffer Object (EBO)

Mesh System

- ☐ Triangle Generation
- ☐ Quad Generation
- ☐ Cube Generation

Vertex Format

Current Vertex:

```
struct Vertex
{
    glm::vec3 Position;
    glm::vec3 Normal;
};
```

Visibility

- ☐ Depth Test
- ☐ Back Face Culling

Lighting

Directional Light

Implemented:

- ☐ Direction
- ☐ Color
- ☐ Intensity

Ambient Light

Implemented:

- ☐ Color

☐ Intensity

Shading

☐ Lambert Diffuse

GLSL

Implemented:

☐ DirectionalLight struct

☐ Uniform communication CPU -> GPU

Current Architecture

Scene

- |— Camera
- |— DirectionalLight
- |— AmbientLight
- └— GameObjects

GameObject

- |— Transform
- └— Components

Components

- └— MeshComponent

Renderer

- |— DrawScene()
- |— DrawGameObject()
- └— DrawMesh()

Systems Not Started

Graphics

☐ Texture System

☐ Texture Loading

☐ UV Coordinates

☐ Materials

☐ Specular Lighting

☐ Model Loading (.obj)

☐ Shadow Mapping

☐ Skybox

☐ PBR

Engine

☐ Asset Manager

☐ Resource Cache

☐ Serialization

☐ Scene Saving

Physics

☐ Collision Detection

☐ Rigid Bodies

Audio

☐ Audio System

Animation

☐ Skeletal Animation

UI

☐ UI Framework

Current Mission

Mission 29

Texture System

Goals:

Add UV coordinates to Vertex.

Understand texture coordinates.

Load images from disk.

Upload textures to GPU.

Sample textures inside fragment shader.